

Report on Internship Project

Summerization of Legal Documents Using NLP

Submitted By

Geethanjali Manoj Kumar (2010096)

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National Institute of Technology Karnataka

as part of the requirements of

Summer Internship [2023]

under the guidance of

Dr. Sowmya Kamath S, Dept of IT, NITK Surathkal

undergone at



HEALTHCARE ANALYTICS AND LANGUAGE ENGINEERING (HALE) LAB

DEPARTMENT OF INFORMATION TECHNOLOGY

NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA, SURATHKAL

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HEALTHCARE ANALYTICS AND LANGUAGE ENGINEERING (HALE) LAB
Dept. of Information Technology
National Institute of Technology Karnataka, Surathkal

C E R T I F I C A T E

This is to certify that the Internship Report entitled **“Summerization of Legal Documents Using NLP”** is submitted by the student mentioned below -

Details of the Intern

Intern Name	Register No.	Branch/Institute	Signature
Geethanjali Manoj Kumar	2010096	B.Tech in IT	Add sign

This report is a record of the work carried out by the Intern as part of the **Summer Internship** during the summer term of the year **2023**, as part of a research project offered by the Healthcare Analytics and Language Engineering (HALE) Lab. The duration of the internship was **4th May 2023** to **14th July 2023**.

Name and Signature of Internship Guide (with date)
Dr. Sowmya Kamath S

DECLARATION

I hereby declare that the project report entitled “**Summarization of Legal Documents Using NLP**” submitted by me as part of the **Summer Internship project** during the **Summer Term of 2023**, is my original work. I declare that the project has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles elsewhere.

Details of the Intern

Intern Name	Register No.	Branch/Institute	Signature
Geethanjali Manoj Kumar	2010096	B.Tech. in IT	Add sign

Place: NITK Surathkal

Date: 09/10/2023

Update project report title

Abstract—As of December 2022 there are 4.3 crore pending cases out of 5 crore total cases. That is a staggering 85 percent of all cases. With the Government also having 50 percent sponsored by the state, India has the most amount of pending legal cases in the world. Automating the initial legal document processing can exponentially expedite the legal process in India.

This paper explores various methods for legal text summarization. Starting with simple extractive summaries by using TF-IDF scores, and then PEGASUS and BERT for creating legal document summaries.

Keywords: legal analytics, NLP, BERT, TF-IDF, PEGASUS

I. INTRODUCTION

A. Challenges

Legal documents are different from those in the open domain in several ways. They are often lengthy and written in formal, complex language. Tasks that are considered relatively simple in the open domain, such as detecting sentence boundaries or Named Entity Recognition (NER), are challenging in the legal domain. Furthermore, legal documents in many countries are unstructured and do not follow any common pattern. They lack demarcations such as headings, etc. Although it is now possible to gain access to unlabelled legal corpora relatively easily, large-scale, task-specific annotated datasets are still difficult to find for many countries/languages. Existing datasets are not well organized, and manual annotation requires the assistance of legal experts, which proves to be expensive both in terms of time and effort. These challenges are discussed in this part.

B. Motivation

As of December 2022 there are 4.3 crore pending cases out of 5 crore total cases. That is a staggering 85 percent of all cases. With the Government also having 50 percent sponsored by the state, India has the most amount of pending legal cases in the world. Automating the initial legal document processing can exponentially expedite the legal process in India.

There is an ardent need to create an automated system to expedite the legal process. This will greatly help the stagnated cases and help bring justice to long pending cases.

II. LITERATURE SURVEY

A. Named Entity Recognition in Indian court judgments

Prathamesh Kalamkar, Astha Agarwal, Aman Tiwari, Smita Gupta³, Saurabh Karn³, Vivek Raghavan¹, 1EkStep Foundation, 2Thoughtworks Technologies India Pvt Ltd., 3Agami

For all NLP based projects we have publicly available standard NERs (Named Entity Recognition), but when it comes to Legal NER there are several unrecognized entities and a separate NER specific to Legal jargon has to be developed. Some sample entities we notice in legal documents are: names of petitioner, respondent, court, statute, provision, precedents, etc. NER is used for information extraction that searches and segmentizes named entities in text. Named entities are defined as proper nouns that refer to specific individuals, places, organizations, or things. NER is a challenging task because named entities can vary in length and can appear in different forms.

The identification of named entities in legal texts is a crucial step in developing other legal Artificial Intelligence applications. Legal named entities are more detailed and nuanced than commonly used named entities like Person, Organization, and Location. In a recent paper, the authors introduced a new corpus of 46,545 annotated legal named entities that are mapped to 14 legal entity types. The paper also describes the development of a baseline model for extracting legally named entities from judgment text.

B. Indian Legal Text Summarization: A Text Normalisation-based Approach

Satyajit Ghosh, Mousumi Dutta, Tanaya Das

The Indian court system has been grappling with the issue of pending cases for a long time. There are over 4 crore cases that are yet to be resolved, and manually summarizing hundreds of documents is a tedious and time-consuming task for legal stakeholders. While many state-of-the-art models for text summarization have emerged as machine learning has progressed, domain-independent models do not perform well with legal texts. Fine-tuning these models for the Indian Legal System is problematic due to a lack of publicly available datasets. To improve the performance of domain-independent models, the authors have proposed a methodology for normalizing legal texts in the Indian context. The authors experimented with two state-of-the-art domain-independent models for legal text summarization, namely BART and PEGASUS. BART and PEGASUS were tested in terms of extractive and abstractive summarization to understand the effectiveness of the text normalization approach. Summarized texts were evaluated by domain experts on multiple parameters using ROUGE metrics, which showed that the proposed text normalization approach is effective in legal texts with domain-independent models

C. Automatic Text Summarization of COVID-19 Medical Research Articles using BERT and GPT-2

Bowen Tan, Virapat Kieuvongngam, Yiming Niu

The COVID-19 pandemic has led to a surge in coronavirus-related literature, which has created a need for the medical community to keep up with the rapidly growing publications. To address this issue, the COVID-19 Open Research Dataset Challenge has released a corpus of scholarly articles and is calling for machine learning approaches to help bridge the gap between researchers and the rapidly growing publications¹²³.

In response to this challenge, researchers have used pre-trained NLP models such as BERT and OpenAI GPT-2 to perform text summarization on this dataset. The results were evaluated using ROUGE scores and visual inspection. The model provides abstractive and comprehensive information based on keywords extracted from the original articles. This work can help the medical community by providing succinct summaries of articles for which the abstracts are not already available¹.

D. Semantic Segmentation of Legal Documents via Rhetorical Roles

Vijit Malik, Rishabh Sanjay, Shouvik Kumar Guha, Angshuman Hazarika, Shubham Nigam, Arnab Bhattacharya, Ashutosh Modi

Legal documents are lengthy, unstructured, and use legal jargon, making them difficult to process automatically using conventional text processing techniques. A legal document processing system would benefit significantly if the documents could be segmented into coherent information units. A new corpus of legal documents has been proposed in this paper, which has been annotated with a set of 13 semantically coherent unit labels (referred to as Rhetorical Roles), such as facts, arguments, statute, issue, precedent, ruling, and ratio. The authors have conducted a thorough analysis of the corpus and the annotations. They have experimented with the task of rhetorical role prediction for automatically segmenting legal documents. Using the created corpus, they have extensively experimented with various deep learning-based baseline models for the task. Furthermore, they have developed a multitask learning (MTL) based deep model with document rhetorical role label shift as an auxiliary task for segmenting a legal document. The proposed model has shown superior performance over existing models. The authors have also experimented with model performance in the case of domain transfer and model distillation techniques to see the model performance in limited data conditions¹.

III. PROBLEM STATEMENT

Summarize legal documents taking into account the complex nature of legal jargon using various NLP models and comparing the results.

IV. RESEARCH OBJECTIVES

- (1) Preprocess and simplify legal documents.
- (2) Identify useful datasets and subsets of the same.

- (3) Create extractive and abstractive summaries of the given dataset.

V. METHODOLOGY AND FRAMEWORK

A. INITIAL WORK

The ILDC dataset is preprocessed by the removal of stop words and stemming. Followed by TF-IDF to find the relevance of tokens after tokenization. This is then followed by a ranking of sentences and a selection of the most relevant sentences to make the extractive summaries.

B. PEGASUS: For Abstractive Summarization

The PEGASUS model is a transformer-based encoder-decoder model that is pre-trained on massive text corpora with a new self-supervised objective called gap-sentences generation (GSG). In this approach, important sentences are removed/masked from an input document and are generated together as one output sequence from the remaining sentences, similar to an extractive summary. The model is designed to perform well on summarization-related downstream tasks. It has been used for legal text summarization and has shown promising results³⁴. In a study conducted on Indian legal documents, the performance of seven machine learning-based summarization models was analyzed. The models that were taken for the analysis were BART, LexRank, TextRank, Luhn, LSA, Legal Pegasus, and Longformer. The study found that Legal Pegasus outperformed all other models in the case of legal summarization.

VI. RESULT AND ANALYSIS

TF-IDF extractive summarization and PEGASUS abstractive summarization are two different approaches to summarizing text. Extractive summarization involves selecting the most important sentences from the original text and forming a summary. On the other hand, abstractive summarization generates new sentences in summary by understanding the text.

TF-IDF extractive summarization works by finding the most important words from the document, then finding sentence scores based on those important words, and finally choosing the most important sentences based on the scores obtained. The approach considers only nouns and verbs to compute the sentence score, as it gives greater accuracy and improves the speed of the algorithm.

PEGASUS abstractive summarization, on the other hand, uses a deep learning model to generate summaries that are more human-like than extractive summaries. It is capable of generating summaries that are not only fluent but also preserve key information and overall meaning. PEGASUS is trained on a large corpus of text data and can generate summaries that are more diverse than extractive summaries.

In general, extractive summarization is faster and easier to implement than abstractive summarization. However, abstractive summarization is better suited for generating summaries that are more human-like and can capture the essence of a document more effectively.

VII. CONCLUSION AND FUTURE WORK

In this paper, I have used TF-IDF for extractive summarization and PEGASUS for abstractive summarization. Based on the results of the research paper, it can be concluded that the proposed methodology for normalizing legal texts can be used to improve the performance of domain-independent models like TF-IDF and PEGASUS.

Based on the research paper titled “Indian Legal Text Summarization: A Text Normalisation-based Approach”, the authors have proposed a methodology for normalizing legal texts in the Indian context. The authors experimented with two state-of-the-art domain-independent models for legal text summarization, namely BART and PEGASUS. BART and PEGASUS are put through their paces in terms of extractive and abstractive summarization to understand the effectiveness of the text normalization approach. Summarized texts are evaluated by domain experts on multiple parameters and using ROUGE metrics. It shows the proposed text normalization approach is effective in legal texts with domain-independent models.

For future work, we can consider experimenting with other domain-independent models like BART, as done in the research paper. You can also explore other techniques for text normalization that are specific to legal texts. Additionally, we can evaluate the performance of the system using ROUGE metrics and by seeking feedback from domain experts.